

Road Traffic Prediction for Munich

By -

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Problem Statement



Preprocessing

Build an end-to-end traffic data pipeline that collects, preprocesses, and transforms raw traffic sensor data into usable features.



Training

Train a predictive model capable of forecasting traffic conditions or congestion from processed traffic features.



Visualize

Deploy the pipeline with a service/API and visualization layer so predictions can be accessed and monitored in real time.





Data Collection

Collect live traffic data from the TomTom API and store it as raw sensor data.



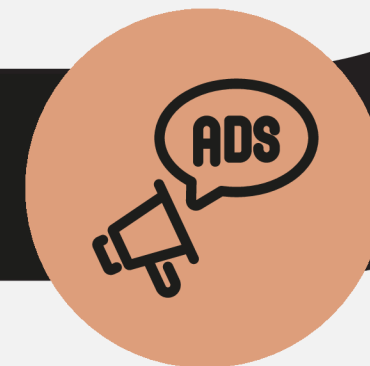
Preprocess

Timestamp parsed as datetime, sorted, removes invalid rows where for eg $\text{current_speed} < 0$, added cyclical features such as `is_peak_morning`.



Deploy

Deploy the model through a FastAPI service and expose results via a Streamlit dashboard, with monitoring support.



Training

Feature set used – cyclical time, peak flags, raw flow, lag / rolling. Target variable is `congestion_level`.
Models evaluated – RandomForest, XGB, LGBM

Pipeline

MLOps Techniques Used

Automated workflows: GitHub Actions run scheduled data collection and nightly retraining pipelines.

Deployment and monitoring: the model is served via FastAPI/Docker, with Prometheus metrics exposed for production monitoring.

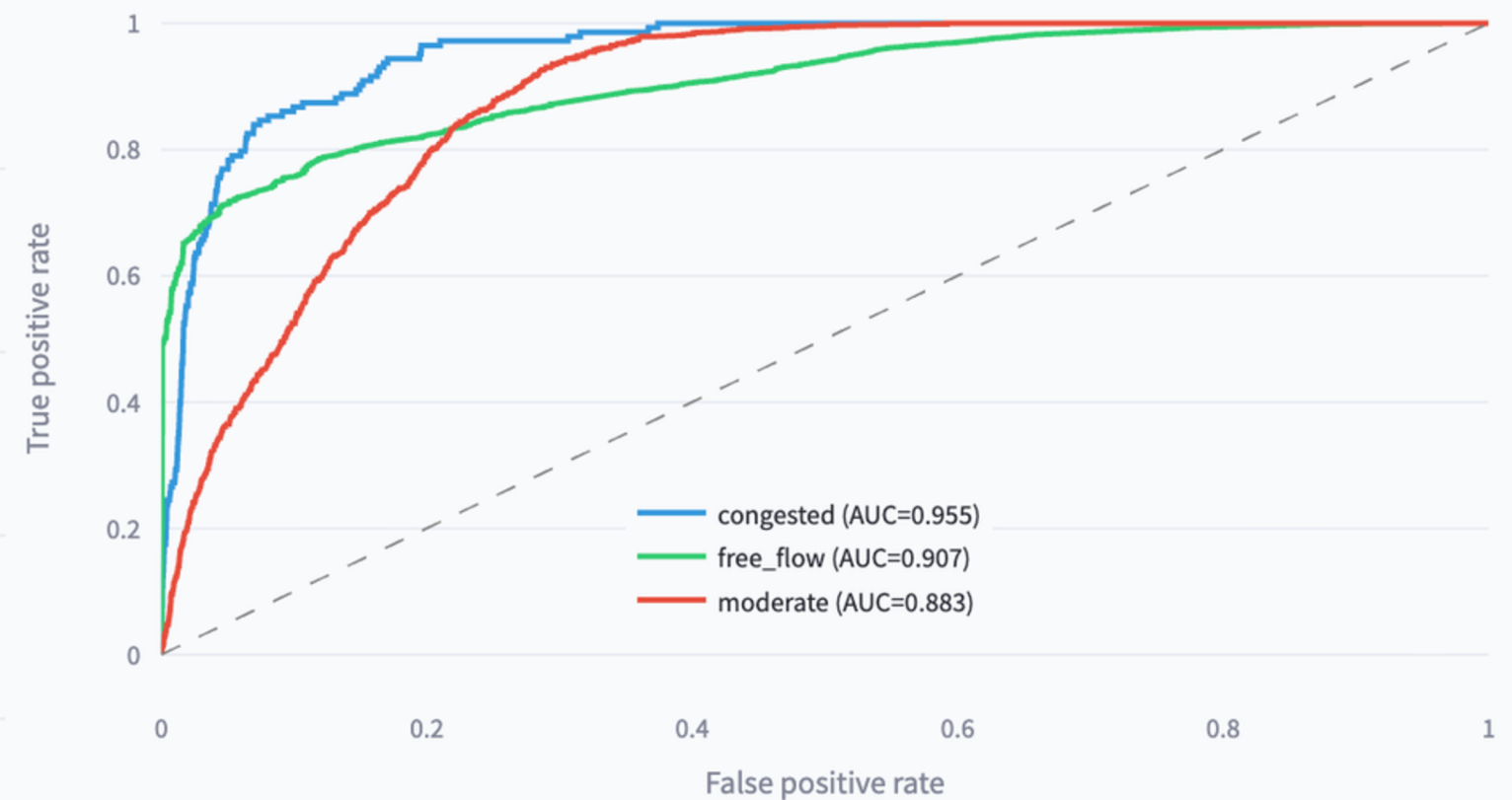
Model tracking and experiment logging: MLflow captures CV metrics, test metrics, plots, and model artifacts.

Model Results

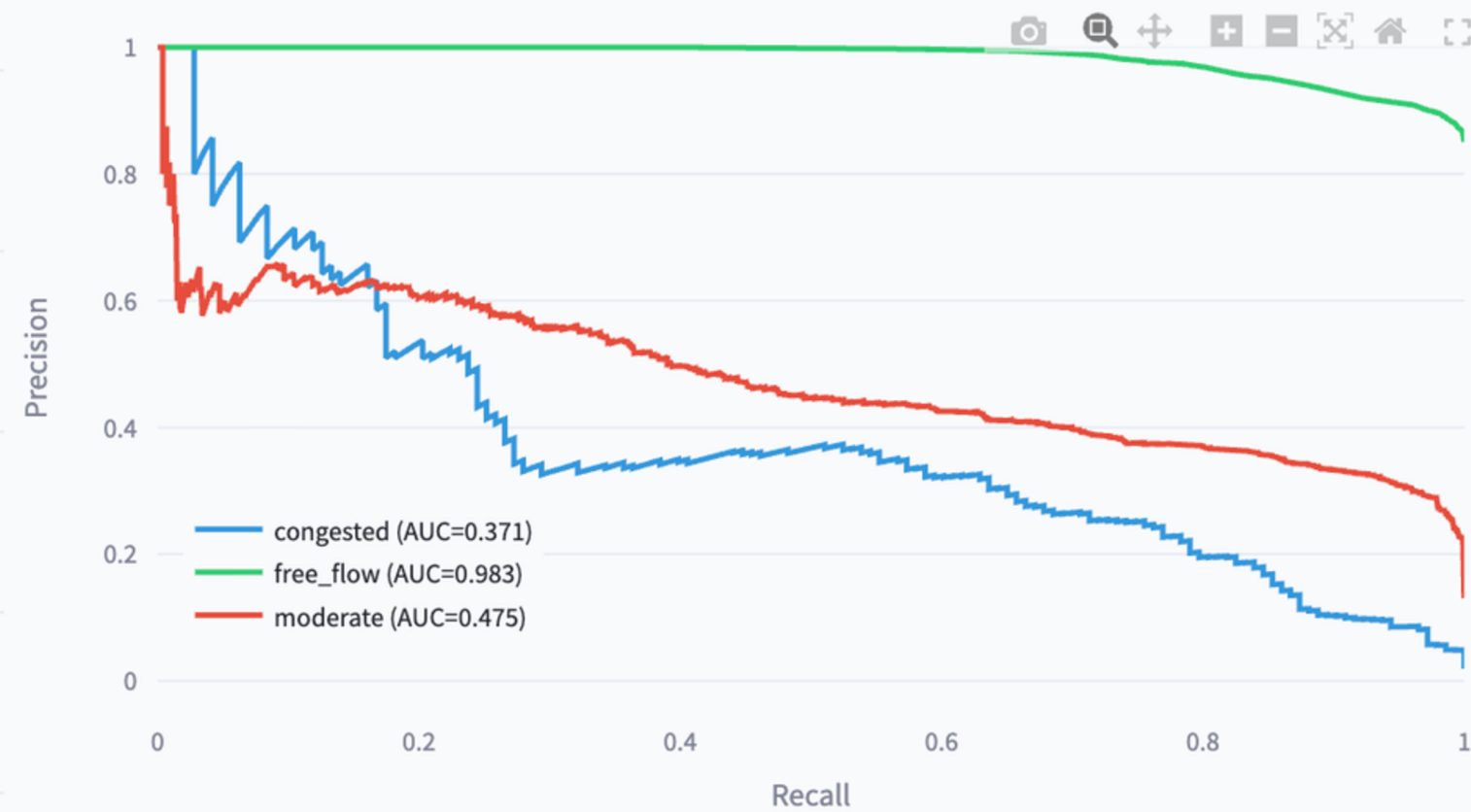
Confusion matrix



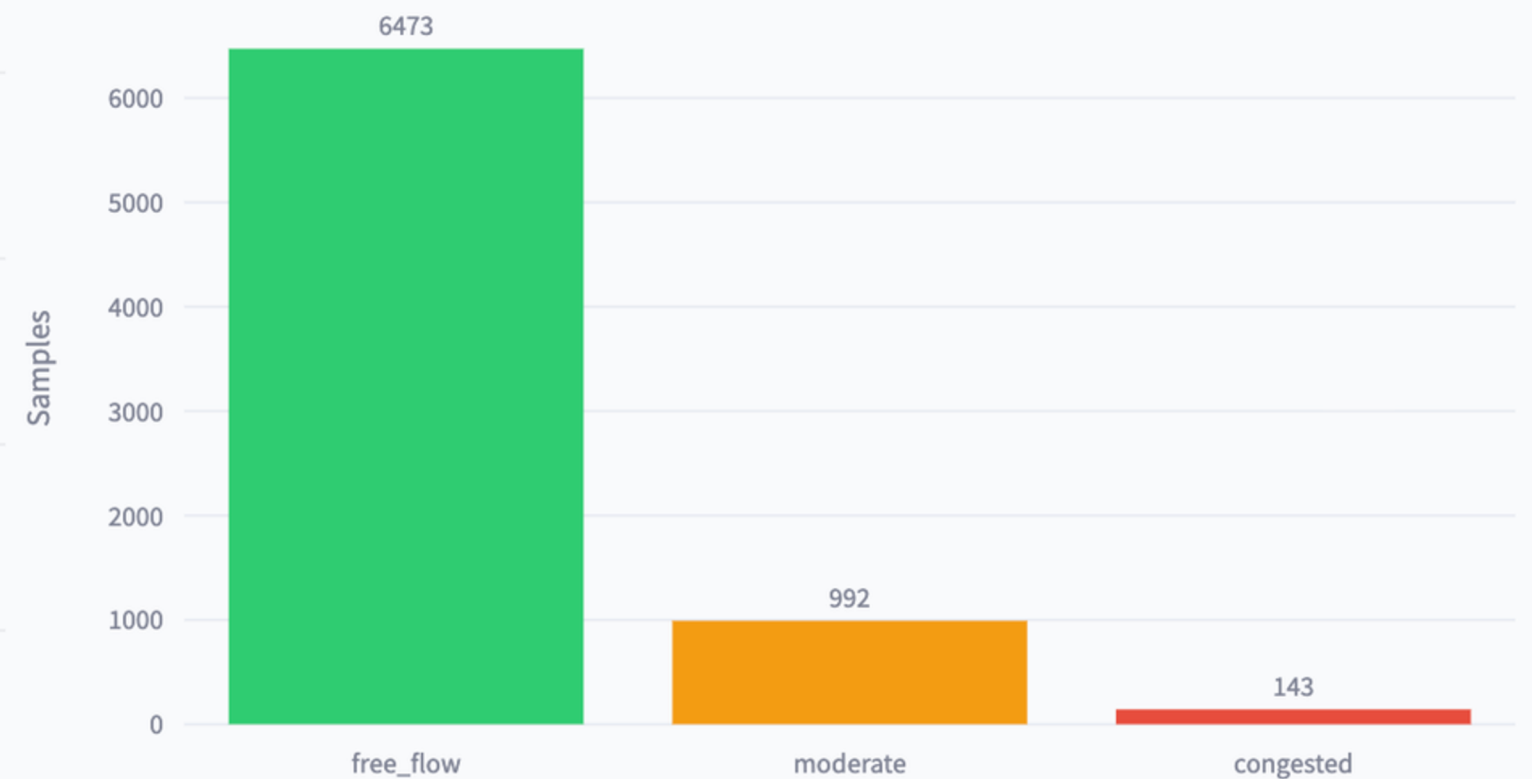
ROC-AUC curve



Precision-recall curve



Class distribution



15-Minute Ahead Prediction — Live Accuracy & Drift

Predictions are made 15 min in advance by `src/evaluate_live.py` (run every 5 min via cron). Each cycle fetches real TomTom data, compares it to the stored prediction, and logs the result here.

At a glance

Latest accuracy

50.0%

↓ -12.5pp

Latest F1 (weighted)

0.4000

Evaluations logged

42

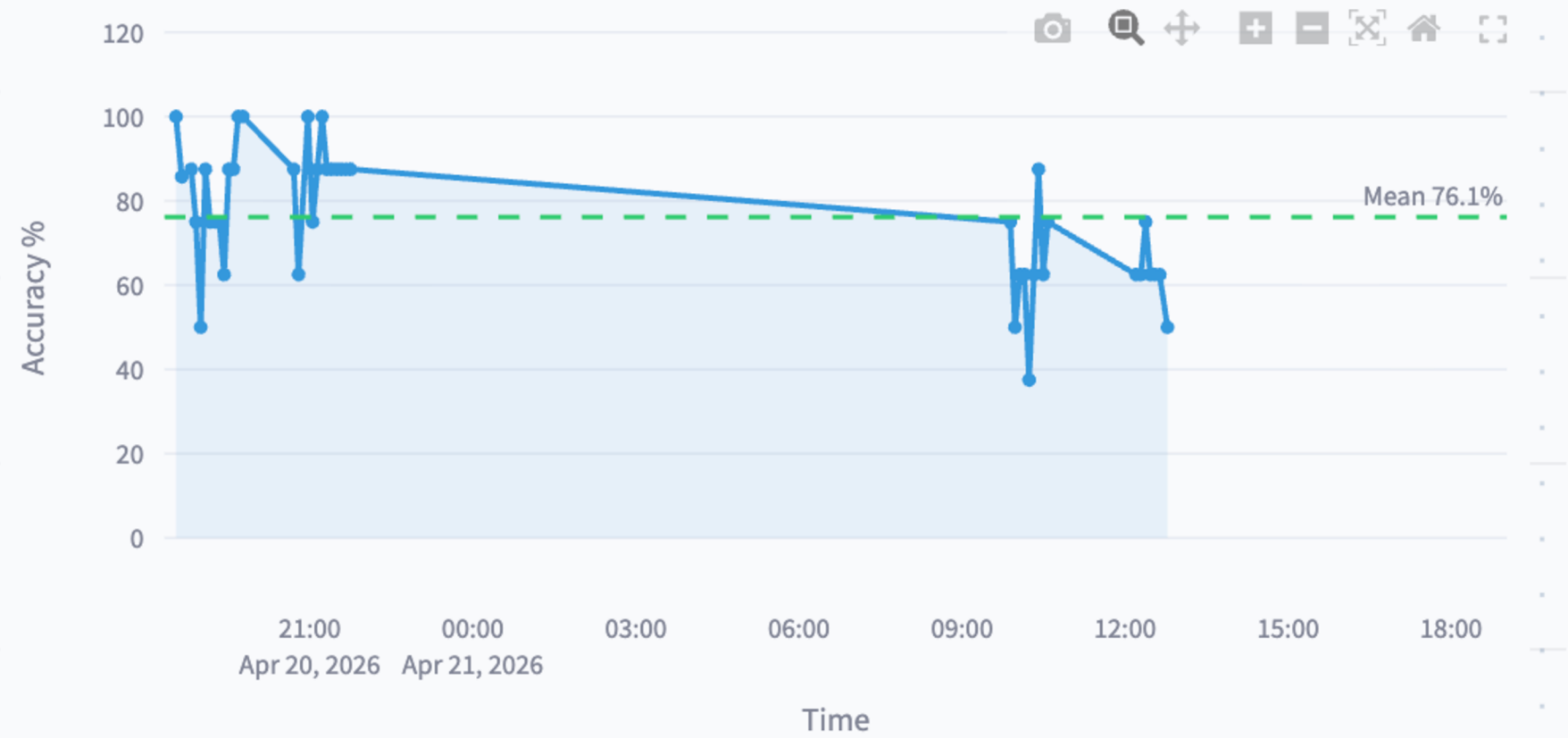
Speed PSI (data drift) ?

0.6569

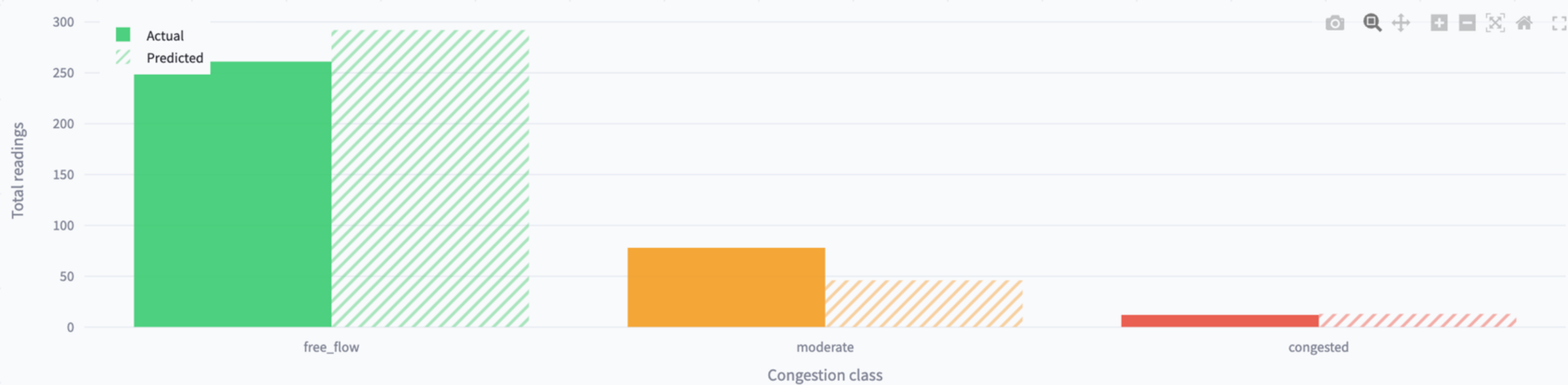
Model confidence drift ?

0.1602

Live accuracy over time (15-min ahead)

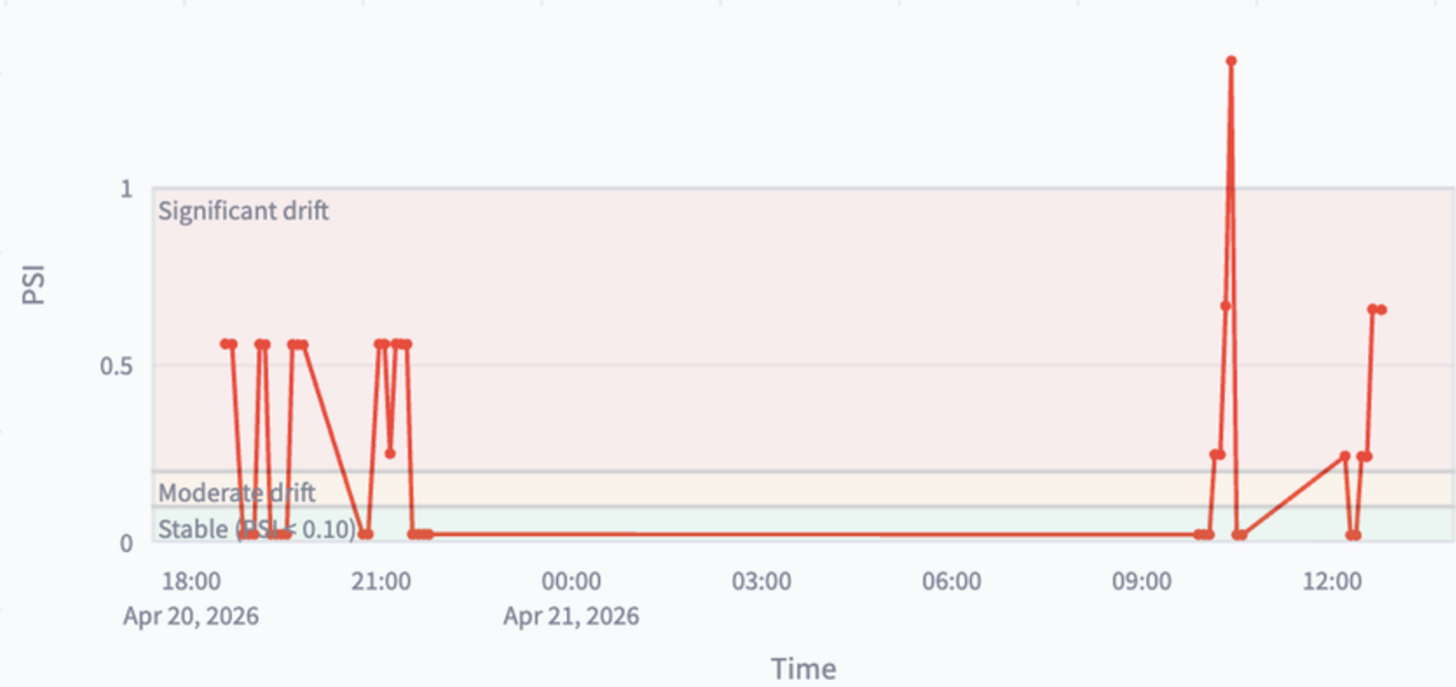


Predicted vs actual label distribution (cumulative)



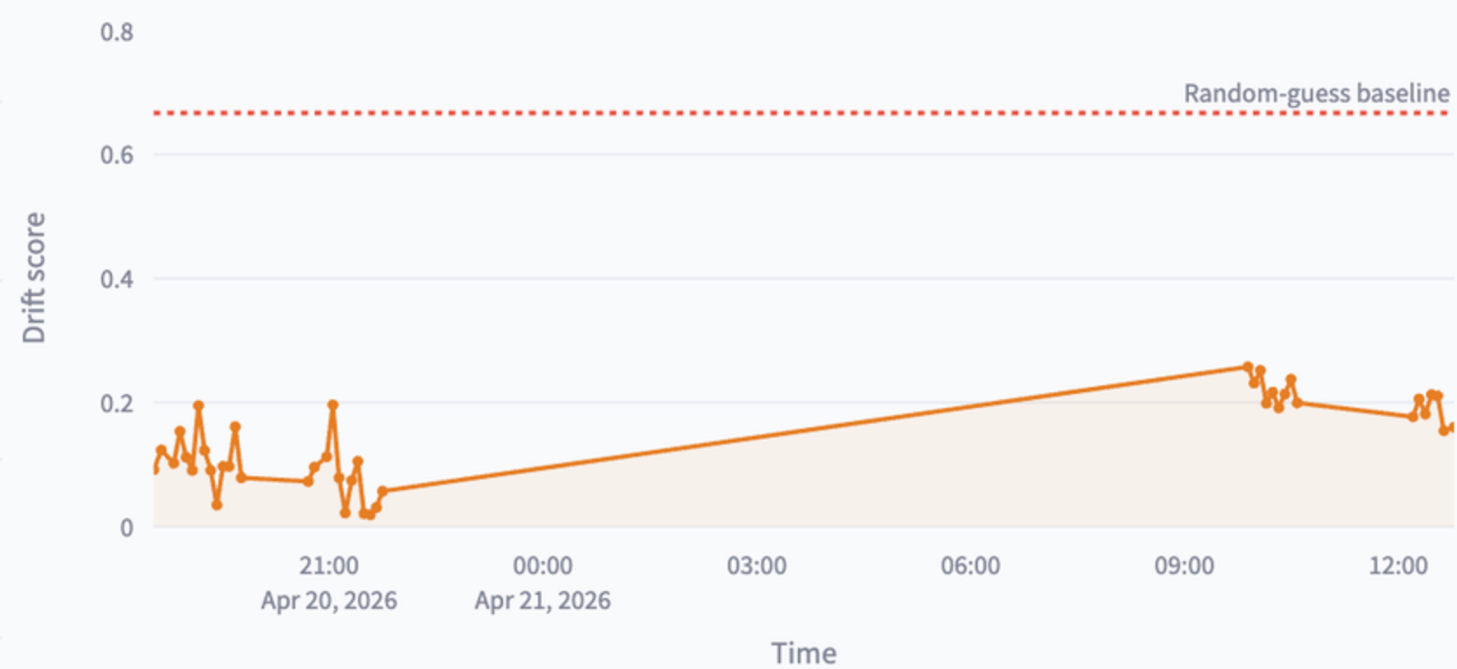
Drift monitoring

Data drift — Speed PSI over time

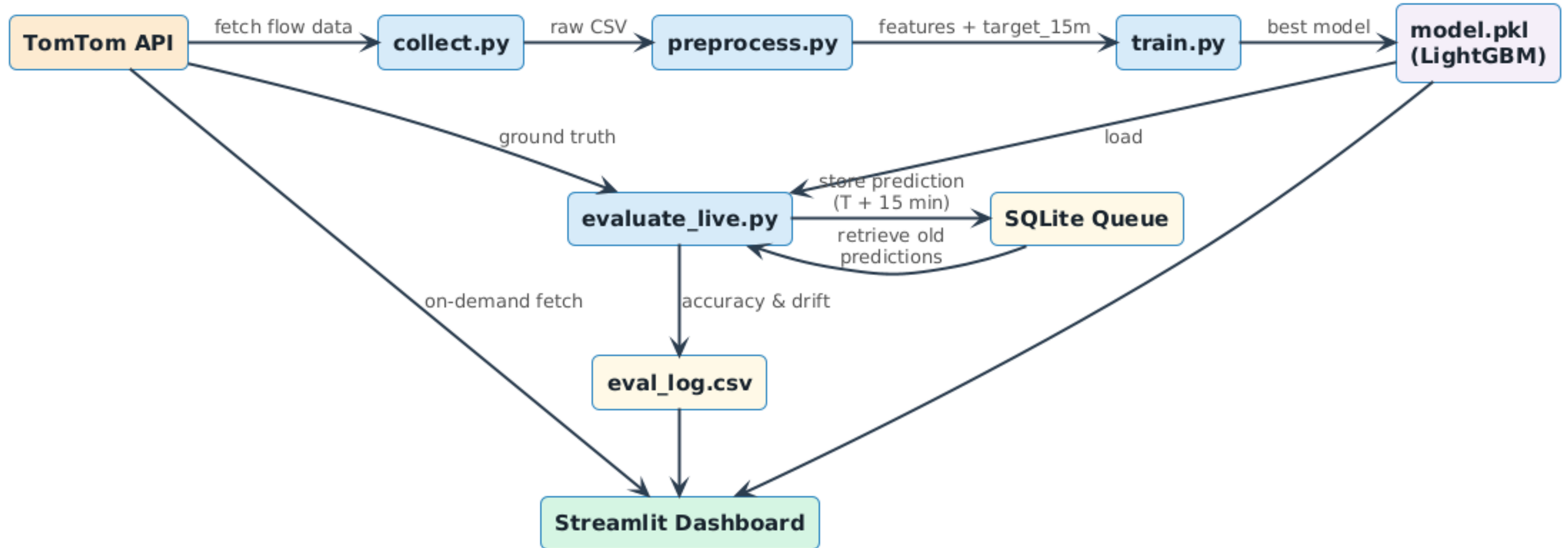


Model drift — Confidence score over time

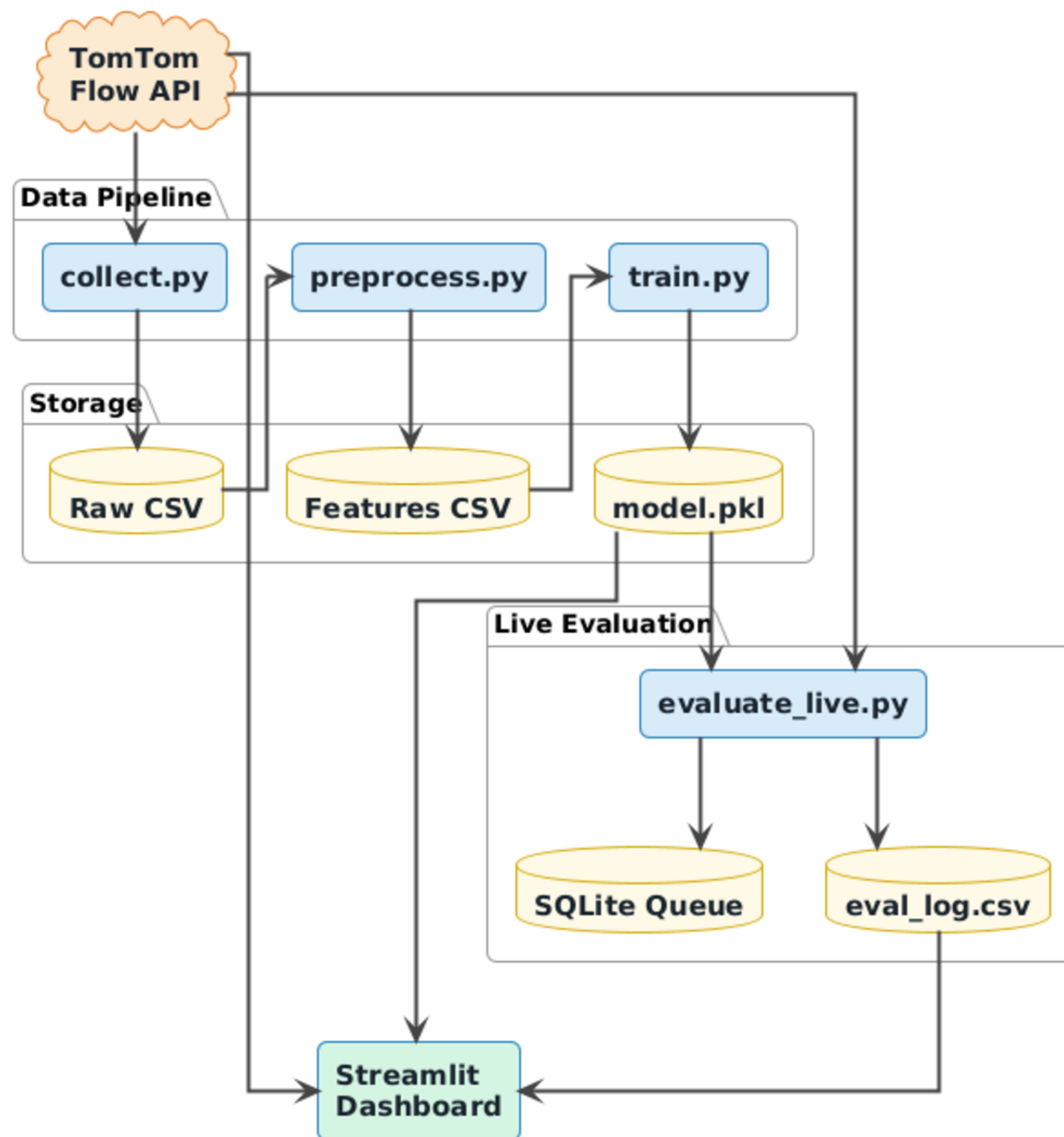
Model drift score = $1 - \text{avg max-class probability}$. A rising score means the model is becoming less certain (possible concept drift).

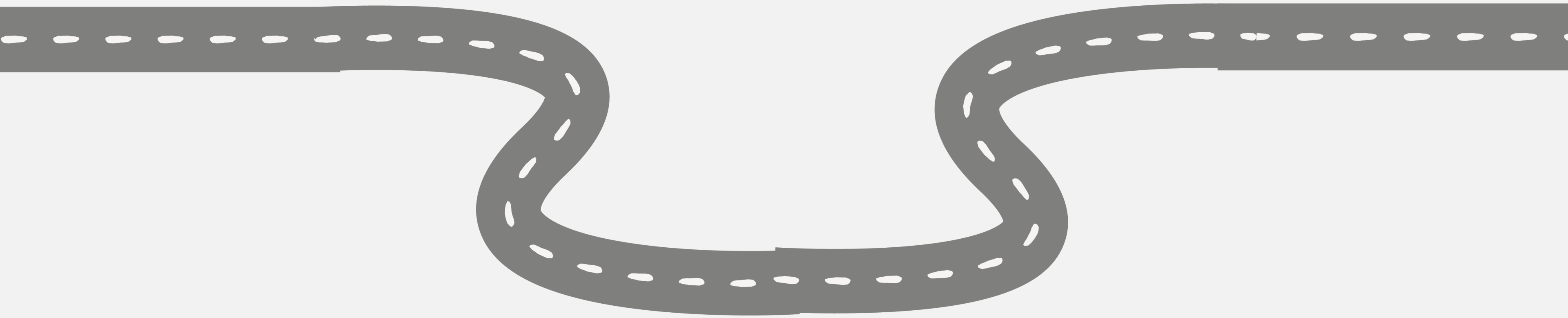


Model Data Flow



System Architecture





Thank You

Github URL Link:

<https://github.com/Sushdi/traffic-pipeline.git>